THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An ovary has 20 ovules and all of them are fertilised. The fruit will have 1

(a) 0 seeds **(b)** 1 seed **(c)** 20 seeds **(d)** 2 seeds

2. Which of the following statements about sex determination is correct? 1

(a) Egg decides the sex **(b)** Sperm decides the sex

(c) Both eggs and sperm always give girls **(d)** Mother's age decides

3. Which of the following transports water and minerals upward? 1

(a) epidermis **(b)** cortex **(c)** phloem **(d)** xylem

4. A person has a swelling in the neck and is lethargic. The cause may be a deficiency of: 1

(a) sodium **(b)** iron **(c)** iodine **(d)** calcium

5. The structure that connects the embryo to the placenta is the 1

(a) urethra **(b)** oviduct **(c)** cervix **(d)** umbilical cord

6. Plants of a clone are grown in different soils and show different heights. This difference is 1

(a) dominant **(b)** a mutation of germ cells

(c) an acquired trait due to environment **(d)** inherited

7. A leaf is kept in the dark for 24 hours and then tested with iodine. It will show: 1

(a) blue-black colour **(b)** red colour

(c) no blue-black colour **(d)** green colour

8. The impulse travels in a neuron from: 1

(a) cell body to dendrite **(b)** axon to dendrite

(c) dendrite to axon end **(d)** synapse to axon

9. A potato tuber has several eyes. Each eye can grow into a plant by 1

(a) vegetative propagation **(b)** budding

(c) regeneration **(d)** spore formation

10. Which of the following genotypes will produce only one type of gamete? 1

(a) Tt **(b)** TT **(c)** Both TT and Tt **(d)** None

11. A person who has run fast feels muscle pain due to: 1

(a) ethanol accumulation **(b)** CO_2 shortage

(c) glucose excess **(d)** lactic acid accumulation

12. The hormone that is present in greater concentration in fruits and seeds is: 1

(a) gibberellin **(b)** ABA **(c)** cytokinin **(d)** auxin

13. Pollen grains of a flower reaching the stigma of another flower on the same plant is 1

(a) self-pollination **(b)** cross-pollination
(c) vegetative propagation **(d)** fertilisation

14. If a human sperm has 22 autosomes and the Y chromosome, the number of chromosomes is 1

(a) 23 **(b)** 22 **(c)** 44 **(d)** 46

15. Loss of water from leaves creates: 1

(a) transpiration pull **(b)** blood pressure
(c) root pressure only **(d)** osmosis only

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the role of the stigma being sticky and feathery in many flowers? 2

17. What fraction of the F2 progeny in a dihybrid cross is round (R_)? 2

18. Explain why the rate of photosynthesis increases with light intensity up to a limit. 2

19. Differentiate between reflex action and voluntary action. 2

20. Why does a bread piece left in a warm moist place get covered with a woolly mass? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A farmer wants a banana plant that is identical to a good plant. Which method should he use and why? 3

22. Show with a Punnett square the cross Rr x Rr for seed shape and give the genotypic and phenotypic ratio. 3

23. Explain the function of valves in the heart and veins. 3

24. Explain why a person who has an increased level of adrenaline appears pale and sweating, with heart pounding. 3

OR

Why are the sperms of mammals motile while eggs are not? 3

25. What is the role of environment in determining traits? Give an example for plants and humans. 3

OR

Explain why diffusion is insufficient for gas transport in large animals. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain digestion in the human mouth, stomach and small intestine with the enzymes and end products. 5

OR

Explain the human brain parts and give two functions of each part, linking to everyday actions. 5

27. A student places Bryophyllum leaves, a rose cutting and a potato eye in moist soil. Explain what happens in each and the type of reproduction involved. 5

OR

A pure tall plant with round seeds is crossed with a dwarf plant with wrinkled seeds. Give the genotype of F1 and the number of each phenotype in F2 out of 160 plants. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND